Lecture 03~04

조합회로 설계

조합회로

- 메모리(또는 상태)를 포함하지 않음
- 같은 입력 → 같은 출력
- Verilog 설계 시 사용 가능 문장
  - assign
  - always @*
    begin
    blocking assignment;
    blocking assignment;
    …
    end

```
a = b
b = a
```

≠

```
a <= b
b <= a
```

blocking assignment → a값 없어짐

non-blocking assignment → a값과 b값 서로 바꿈

연산 공유(operator sharing)

- 0.55-μm CMOS standard-cell technology

비트 크기	자주 쓰는 연산자									
	nand	xor	$>_a$	$>_d$	=	$+1_a$	$+1_d$	$+_a$	$+_d$	MUX
8	8	22	25	68	26	27	33	51	118	21
16	16	44	52	102	51	55	73	101	265	42
32	32	85	105	211	102	113	153	203	437	85
64	64	171	212	398	204	227	313	405	755	171
8	0.1	0.4	4.0	1.9	1.0	2.4	1.5	4.2	3.2	0.3
16	0.1	0.4	8.6	3.7	1.7	5.5	3.3	8.2	5.5	0.3
32	0.1	0.4	17.6	6.7	1.8	11.6	7.5	16.2	11.1	0.3
64	0.1	0.4	35.7	14.3	2.2	24.0	15.7	32.2	22.9	0.3

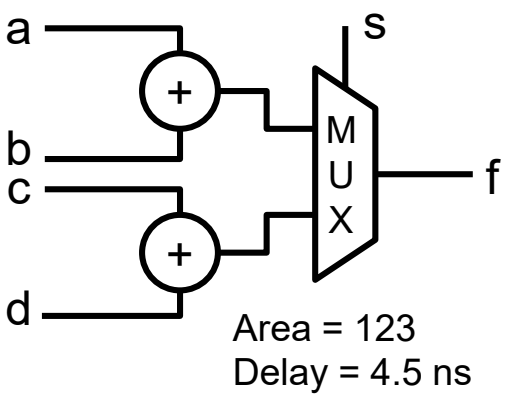
Area (gate count)

Delay (ns)

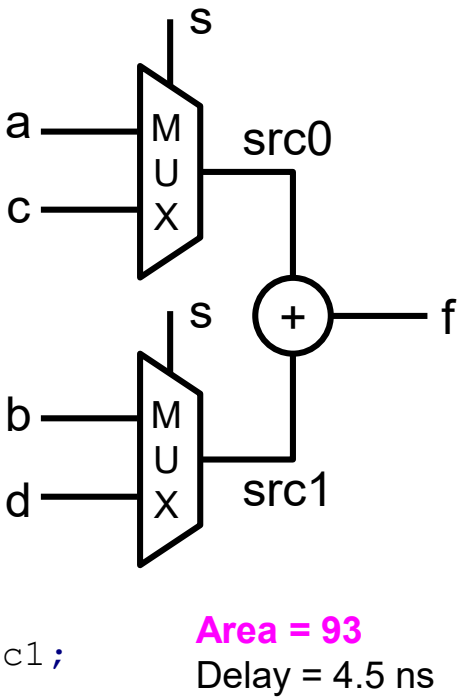
연산 공유(operator sharing)

- 같은 연산을 여러 번 사용하면 → 연산 공유
  - 회로 크기 줄임
  - 전파 지연 시간(propagation delay time) 줄일 수 있음

```
if (s == 1)
    f = a + b
else
    f = c + d
```



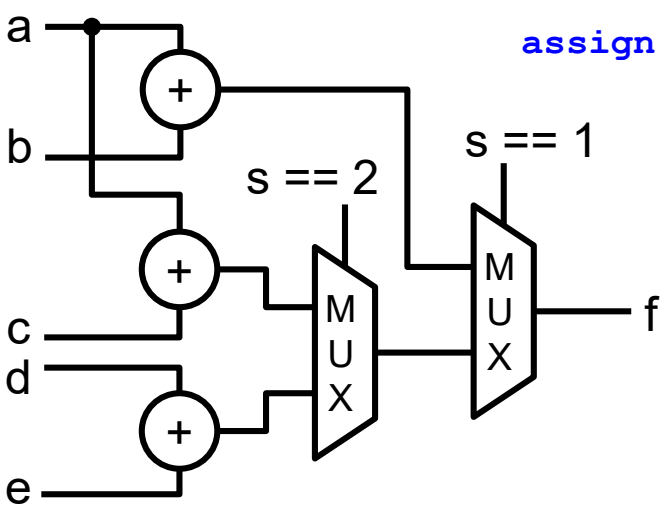

```
if (s == 1) begin
    src0 = a;
    src1 = b;
end
else begin
    src0 = c;
    src1 = d;
end

assign f = src0 + src1;
```



연산 공유(operator sharing)

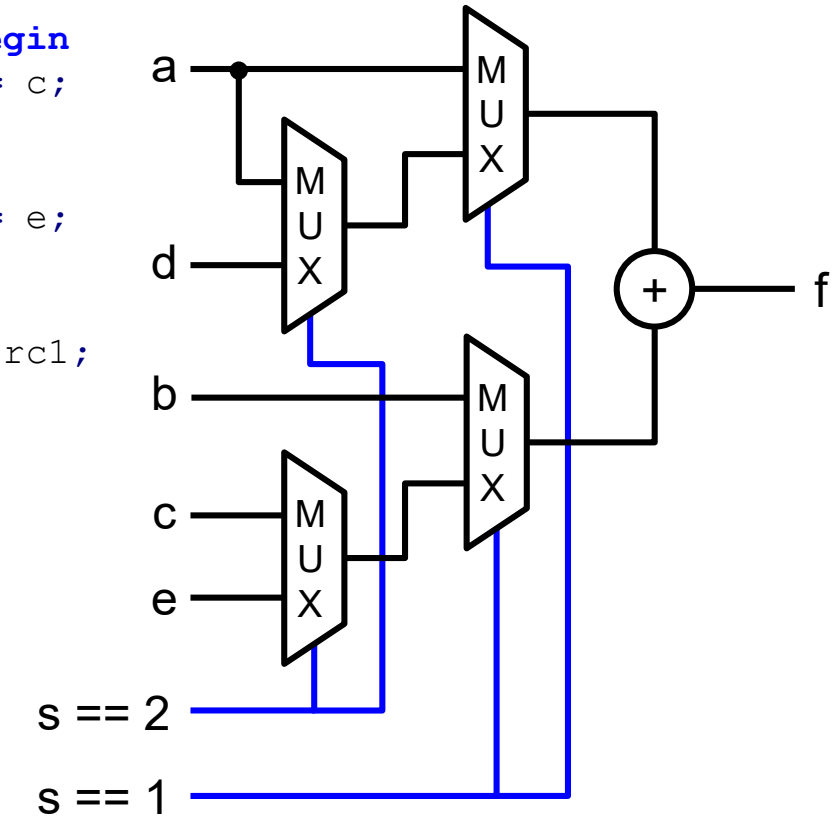
```
if (s == 1)
    f = a + b;
else if (s == 2)
    f = a + c;
else
    f = d + e;
```

Area = 195
Delay = 4.8 ns




```
if (s == 1) begin
    src0 = a; src1 = b;
end
else if (s == 2) begin
    src0 = a; src1 = c;
end
else begin
    src0 = d; src1 = e;
end

assign f = src0 + src1;
```

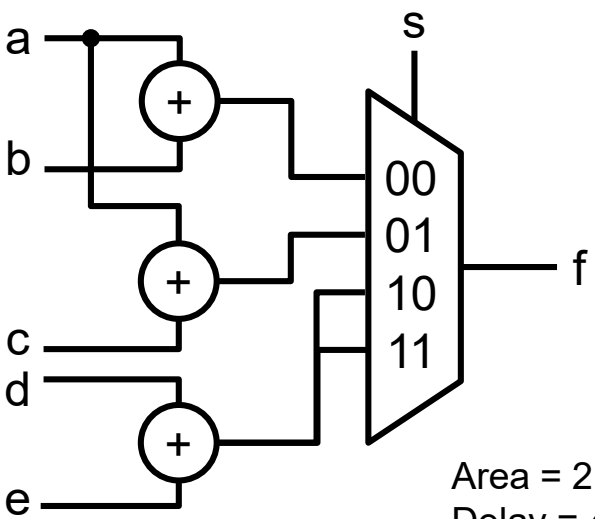
Area = 135
Delay = 4.8 ns



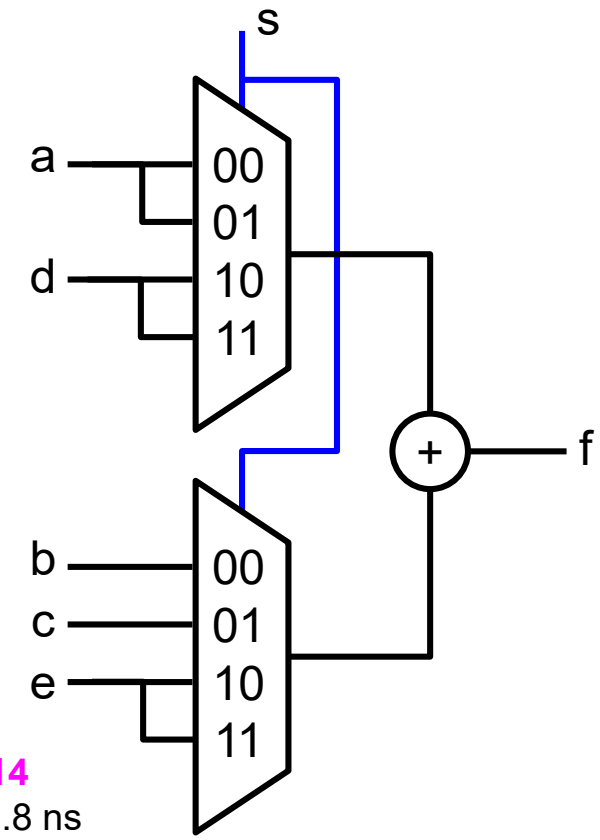
연산 공유(operator sharing)

```
case (s)
   2'b00: f = a + b;
   2'b01: f = a + c;
   default: f = d + e;
endcase
```

```
case (s)
   2'b00: begin
      src0 = a; src1 = b;
   end
   2'b01: begin
      src0 = a; src1 = c;
   end
   default: begin
      src0 = d; src1 = e;
   end
endcase

assign f = src0 + src1;
```



Area = 216
Delay = 4.8 ns



Area = 114
Delay = 4.8 ns

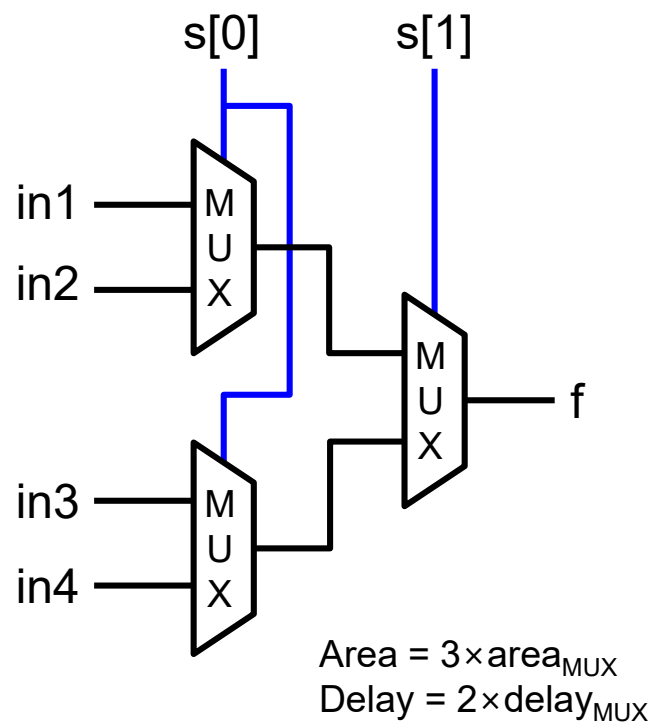
연산 공유(operator sharing)

- 4-to-1 MUX의 area 및 delay 계산?

s[1]	s[0]	f
0	0	in1
0	1	in2
1	0	in3
1	1	in4

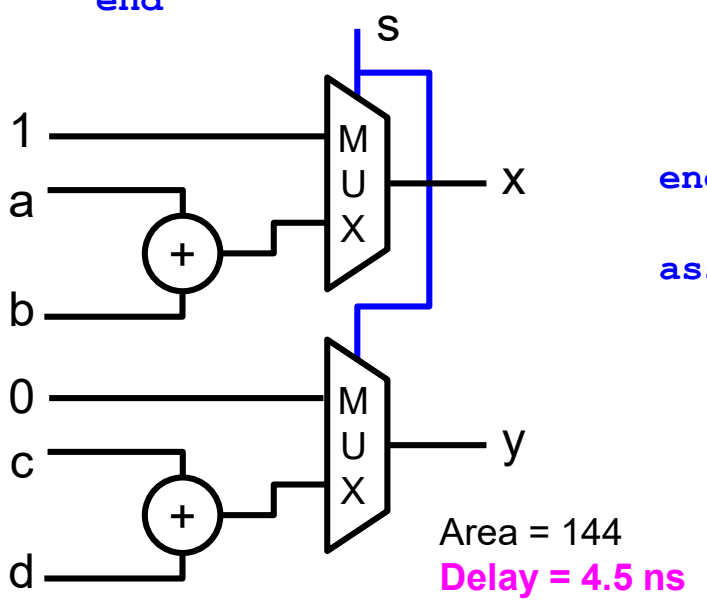
2-to-1 MUX

s[1]	f
0	MUX1의 출력
1	MUX2의 출력



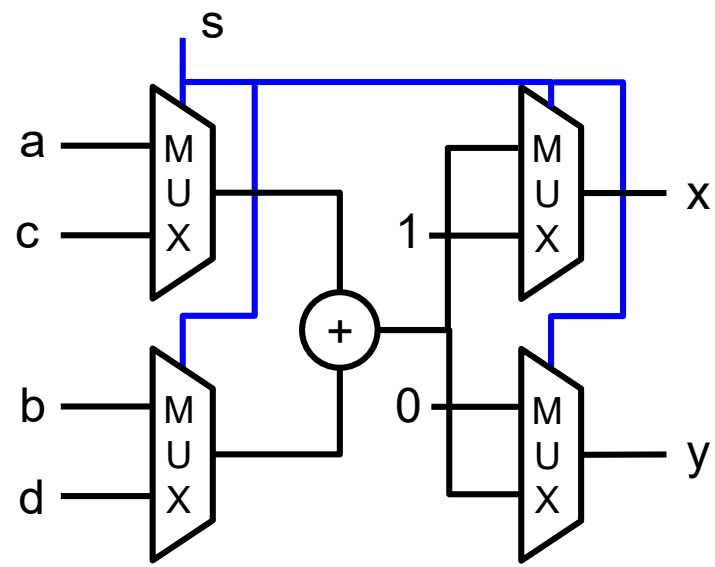
연산 공유(operator sharing)

```
if s begin
    x = a + b;
    y = 0;
end
else begin
    x = 1;
    y = c + d;
end
```



Area = 144
Delay = 4.5 ns

```
if s begin
    src0 = a;
    src1 = b;
    x = sum;
    y = 0;
end
else begin
    src0 = c;
    src1 = d;
    x = 1;
    y = sum;
end

assign sum = src0 + src1;
```



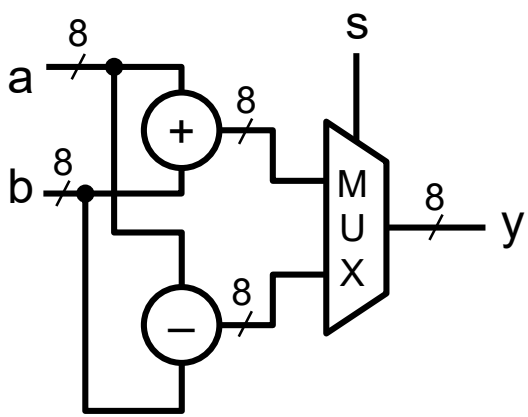
Area = 135
Delay = 4.8 ns

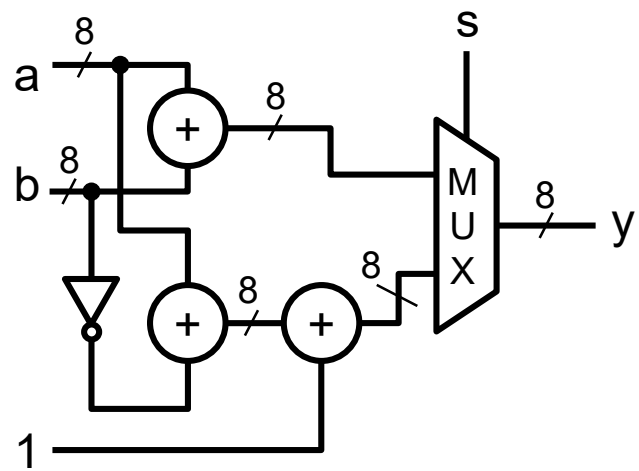
이 경우는 연산 공유 할 필요 없음!

함수 공유(function sharing)

s	f
0	a + b
1	a – b

s	f
0	a + b
1	a – b = a + (~b + 1)

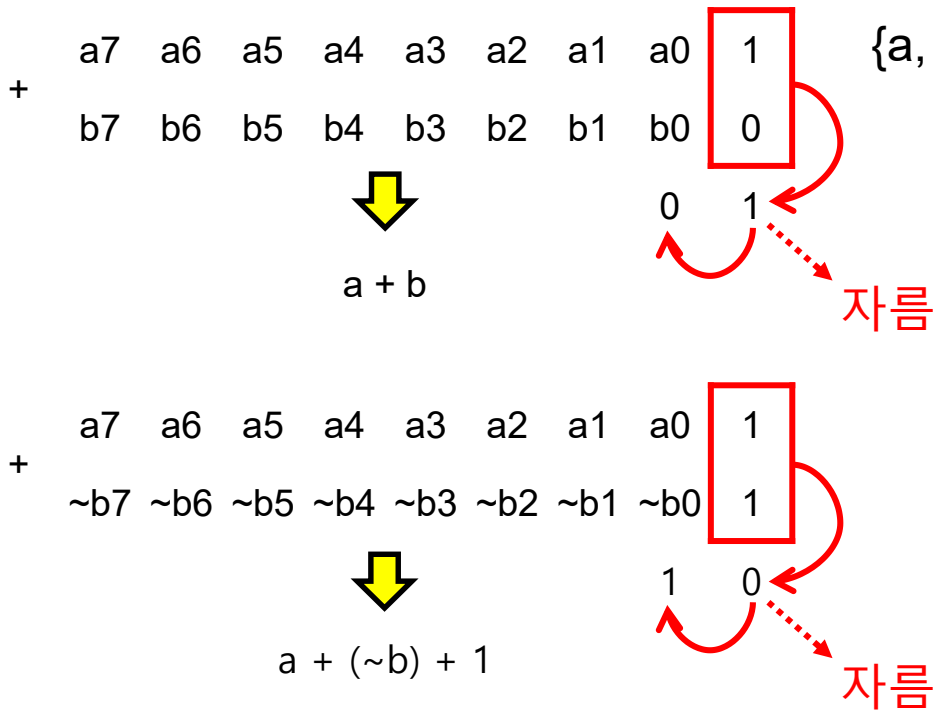


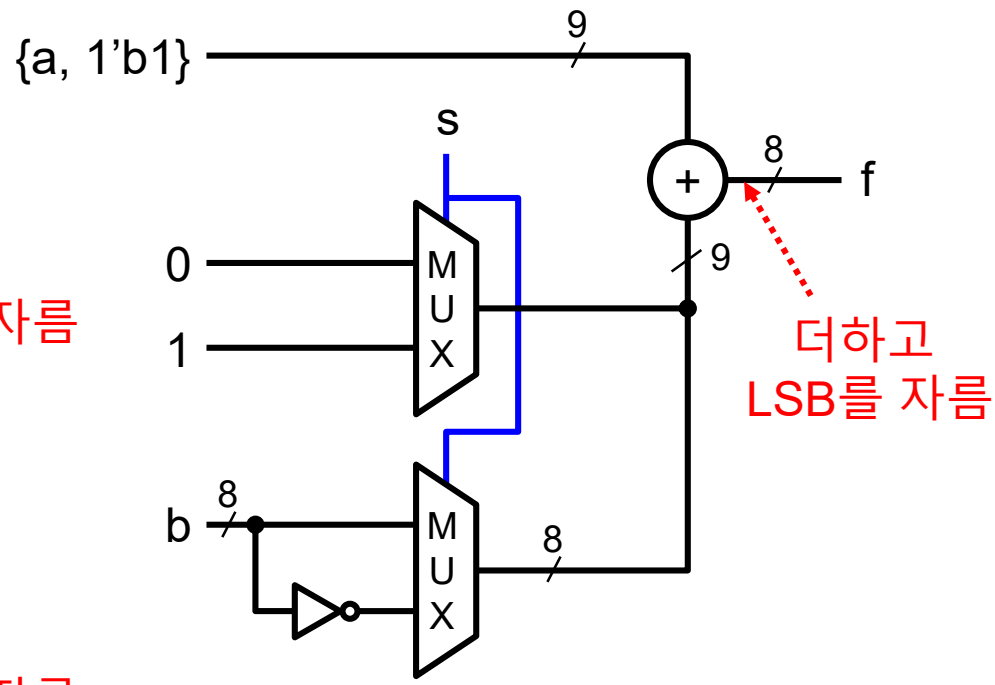


더하기 개수를 줄인 방법?

함수 공유(function sharing)

- 더하기의 carry 비트 활용





Full comparator

- 효율적인 full comparator

비트 크기	관계 연산자		
	$>_a$	$>_d$	=
8	25	68	26
16	52	102	51
32	105	211	102
64	212	398	204
8	4.0	1.9	1.0
16	8.6	3.7	1.7
32	17.6	6.7	1.8
64	35.7	14.3	2.2

Area (gate count)

Delay (ns)

단순한 설계

```
assign gt = (a > b) ? 1 : 0;
assign lt = (a < b) ? 1 : 0;
assign eq = (a == b) ? 1 : 0;
```

효율적인 설계

```
assign gt = (a > b) ? 1 : 0;
assign eq = (a == b) ? 1 : 0;
assign lt = ~(gt | eq);
```

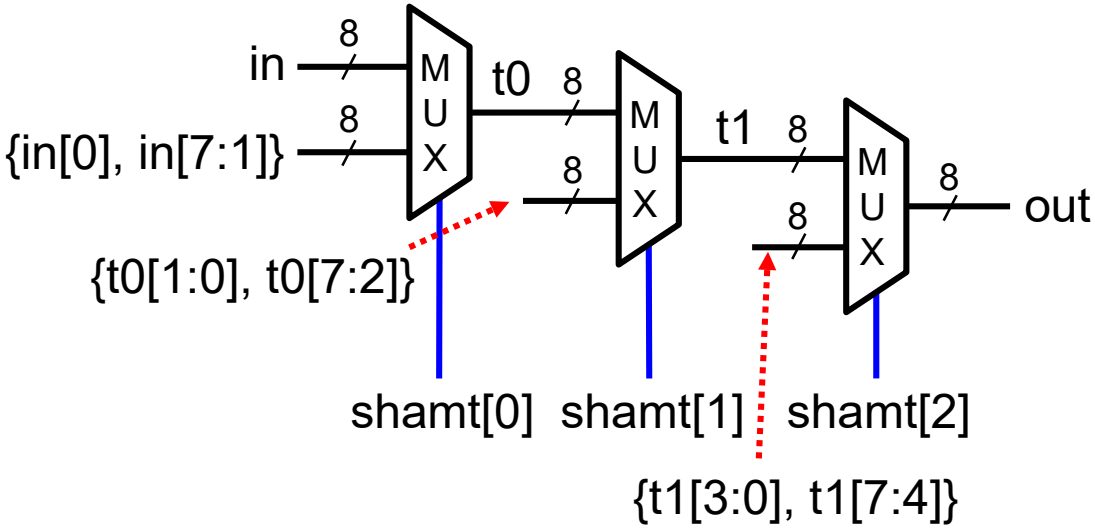
Barrel shifter

- 기본 시프트 연산자 (>>, <<, >>>, <<<)
  - Rotation 불가능
- Barrel 시프트
  - Rotation 가능
  - 입출력 포트
    - 입력: in 및 shamt(시프트 량)
    - 출력: out
  - 동작 원리
    - shamt의 비트 크기에 따라 시프트 단계로 나눔
    - 예: 3-bit shamt = shamt[2]*2^2 + shamt[1]*2^1 + shamt[0]*2^0

→ 첫 단계: 시프트 하지 않거나 1비트(2^0) 시프트
둘 단계: 시프트 하지 않거나 2비트(2^1) 시프트
셋 단계: 시프트 하지 않거나 4비트(2^2) 시프트

Barrel shifter

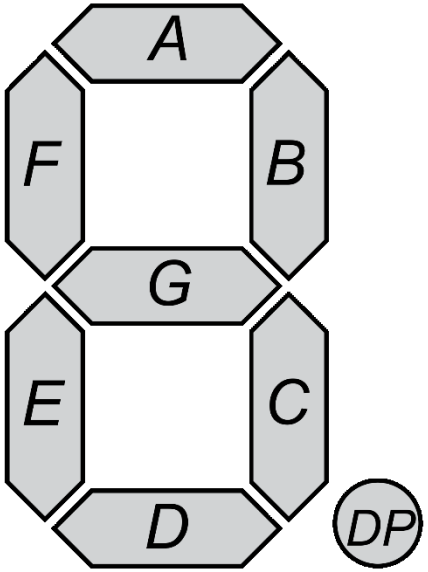
- 코드




```
module barrel_shifter(
    input  [7:0] in,
    input  [2:0] shamt,
    output [7:0] out
);

wire [7:0] t0, t1;

assign t0  = shamt[0] ? {in[0],   in[7:1]} : in; // 1st stage
assign t1  = shamt[1] ? {t0[1:0], t0[7:2]} : t0; // 2nd stage
assign out = shamt[2] ? {t2[3:0], t1[7:4]} : t1; // 3rd stage

endmodule
```

7-segment LED decoder

- 7-segment LED 동작 원리



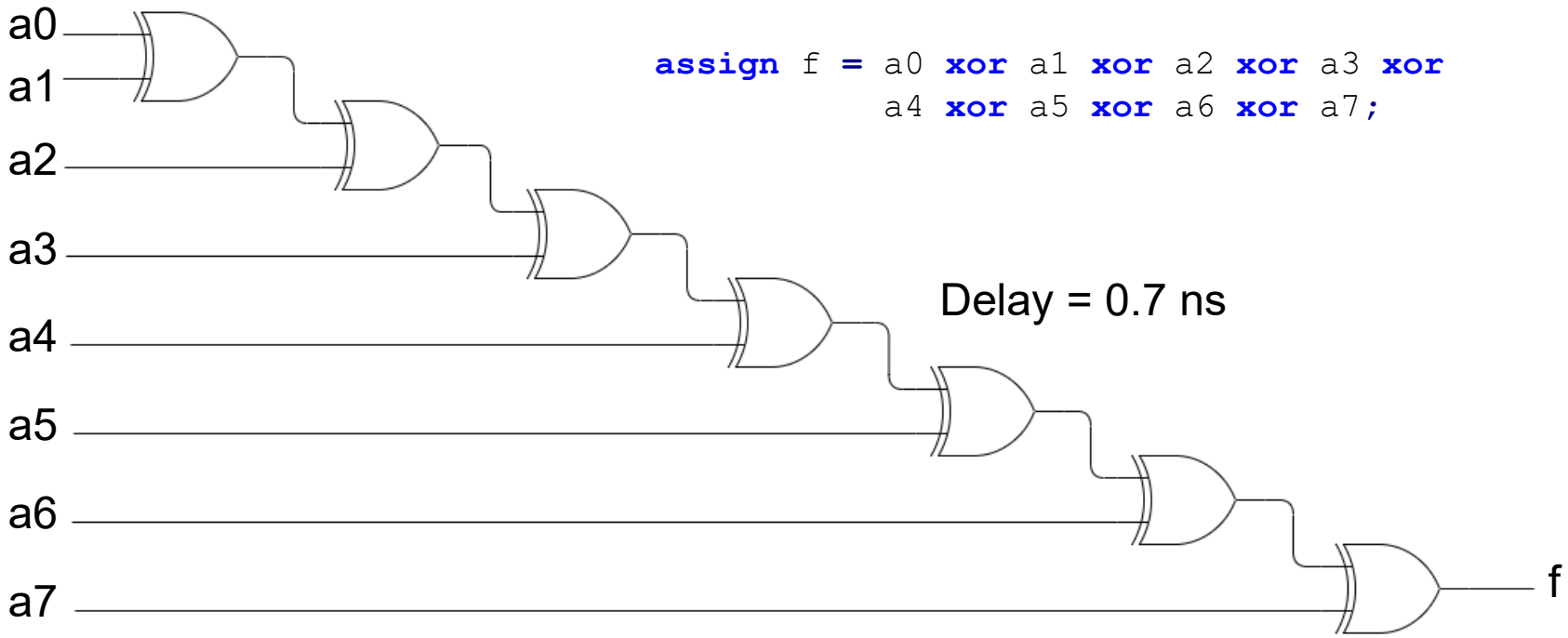
숫자	Active high							Active low						
	A	B	C	D	E	F	G	A	B	C	D	E	F	G
0	1	1	1	1	1	1	0	0	0	0	0	0	0	1
1	0	1	1	0	0	0	0	1	0	0	1	1	1	1
2	1	1	0	1	1	0	1	0	0	1	0	0	1	0
3	1	1	1	1	0	0	1	0	0	0	0	1	1	0
4	0	1	1	0	0	1	1	1	0	0	1	1	0	0
5	1	0	1	1	0	1	1	0	1	0	0	1	0	0
6	1	0	1	1	1	1	1	0	1	0	0	0	0	0
7	1	1	1	0	0	0	0	0	0	0	1	1	1	1
8	1	1	1	1	1	1	1	0	0	0	0	0	0	0
9	1	1	1	1	0	1	1	0	0	0	0	1	0	0

7-segment LED decoder

```
module sseg_decoder(
   input       [3:0] num,
   input             dp,
   output reg [7:0] sseg
);

always @* begin
   case (num)
      4'd0: sseg[6:0] = 7'b111_1110;
      4'd1: sseg[6:0] = 7'b011_0000;
      4'd2: sseg[6:0] = 7'b110_1101;
      4'd3: sseg[6:0] = 7'b111_1001;
      4'd4: sseg[6:0] = 7'b011_0011;
      4'd5: sseg[6:0] = 7'b101_1011;
      4'd6: sseg[6:0] = 7'b101_1111;
      4'd7: sseg[6:0] = 7'b111_0000;
      4'd8: sseg[6:0] = 7'b111_1111;
      default: sseg[6:0] = 7'b111_1011; // 4’d9
   endcase
   sseg[7] = dp;
end

endmodule
```

레이아웃 위한 설계

- 괄호 사용 → 회로 레이아웃 조절 가능
- 예: xor 회로




```
assign f = a0 xor a1 xor a2 xor a3 xor
           a4 xor a5 xor a6 xor a7;
```

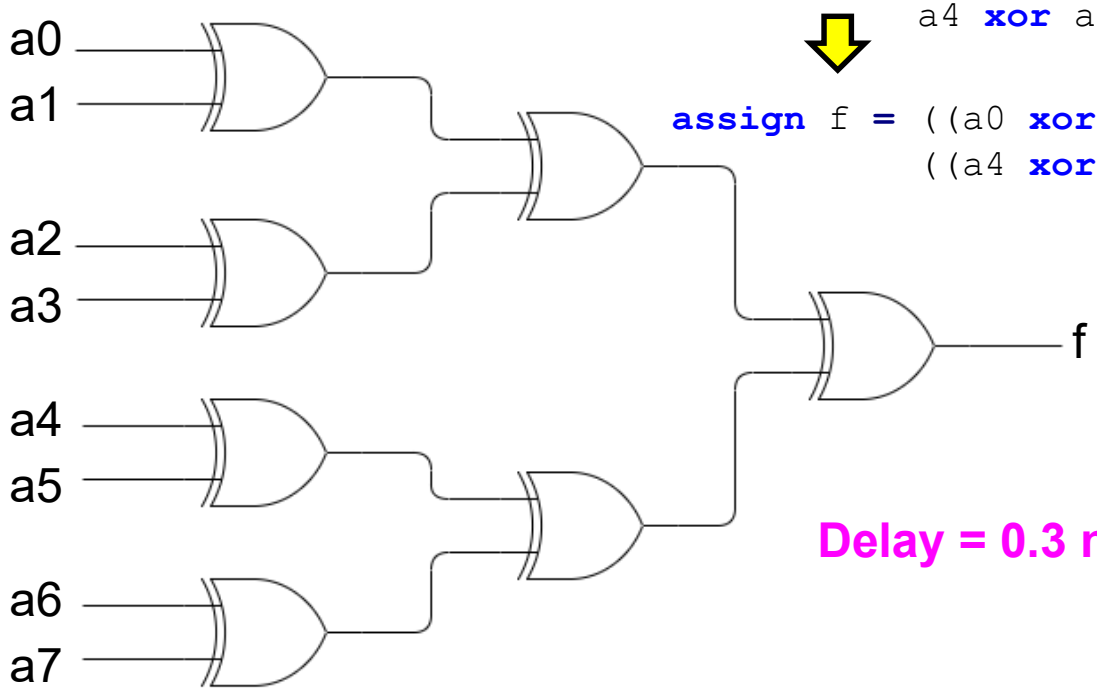
레이아웃 위한 설계

- 괄호 사용 → 회로 레이아웃 조절 가능
- 예: xor 회로

```
assign f = a0 xor a1 xor a2 xor a3 xor
           a4 xor a5 xor a6 xor a7;
```

⇩

```
assign f = ((a0 xor a1) xor (a2 xor a3)) xor
           ((a4 xor a5) xor (a6 xor a7));
```



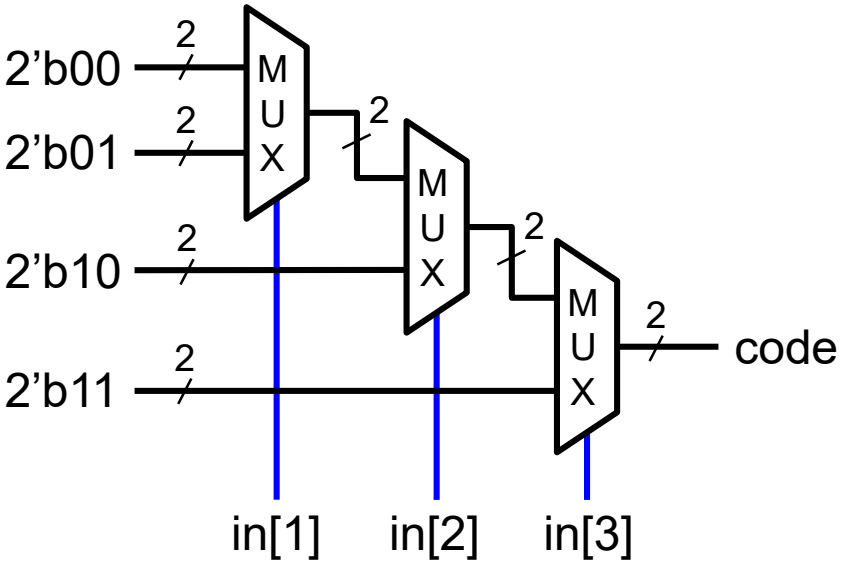
Delay = 0.3 ns

레이아웃 위한 설계

- 예: 4-to-2 priority encoder

in	code
1xxx	11
01xx	10
001x	01
0001	00

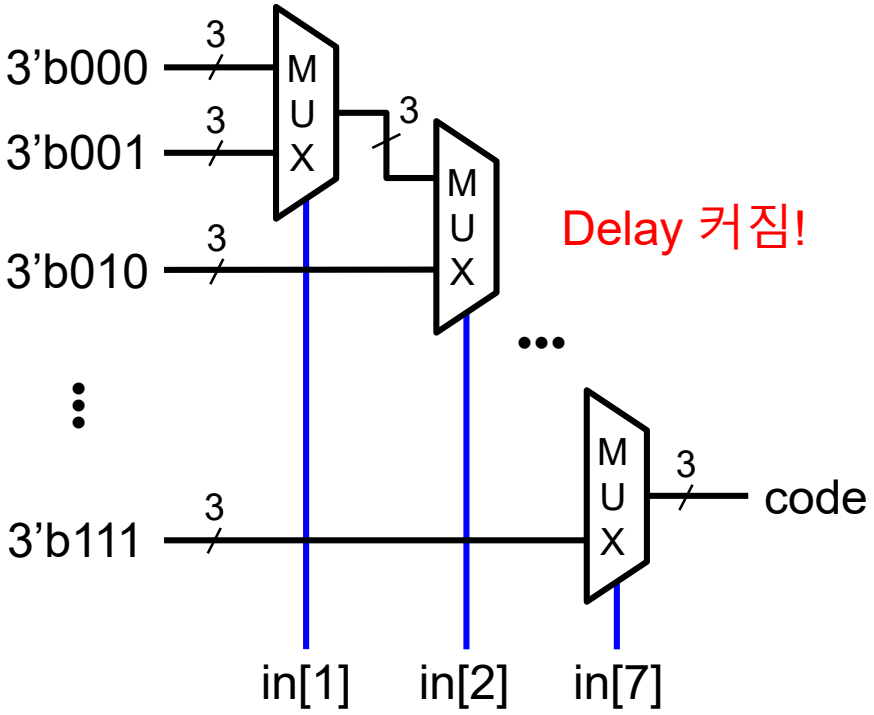
```
module pri_encoder(
   input  [3:0] in,
   output [1:0] code
);

assign code = (in[3]) ? 2'b11 :
              (in[2]) ? 2'b10 :
              (in[1]) ? 2'b01 :
                        2'b00;

endmodule
```



레이아웃 위한 설계

- 예: 8-to-3 priority encoder

```
module pri_encoder83(
    input  [7:0] in,
    output [2:0] code
);

assign code = (in[7]) ? 3'b111 :
              (in[6]) ? 3'b110 :
              (in[5]) ? 3'b101 :
              (in[4]) ? 3'b100 :
              (in[3]) ? 3'b011 :
              (in[2]) ? 3'b010 :
              (in[1]) ? 3'b001 :
                        3'b000;

endmodule
```



레이아웃 위한 설계

- 예: 4-to-2 priority encoder로 8-to-3 priority encoder

in	code
1xxx xxxx	111
01xx xxxx	110
001x xxxx	101
0001 xxxx	100
0000 1xxx	011
0000 01xx	010
0000 001x	001
0000 0001	000

in	code
1xxx xxxx	111
01xx xxxx	110
001x xxxx	101
0001 xxxx	100
0000 1xxx	011
0000 01xx	010
0000 001x	001
0000 0001	000

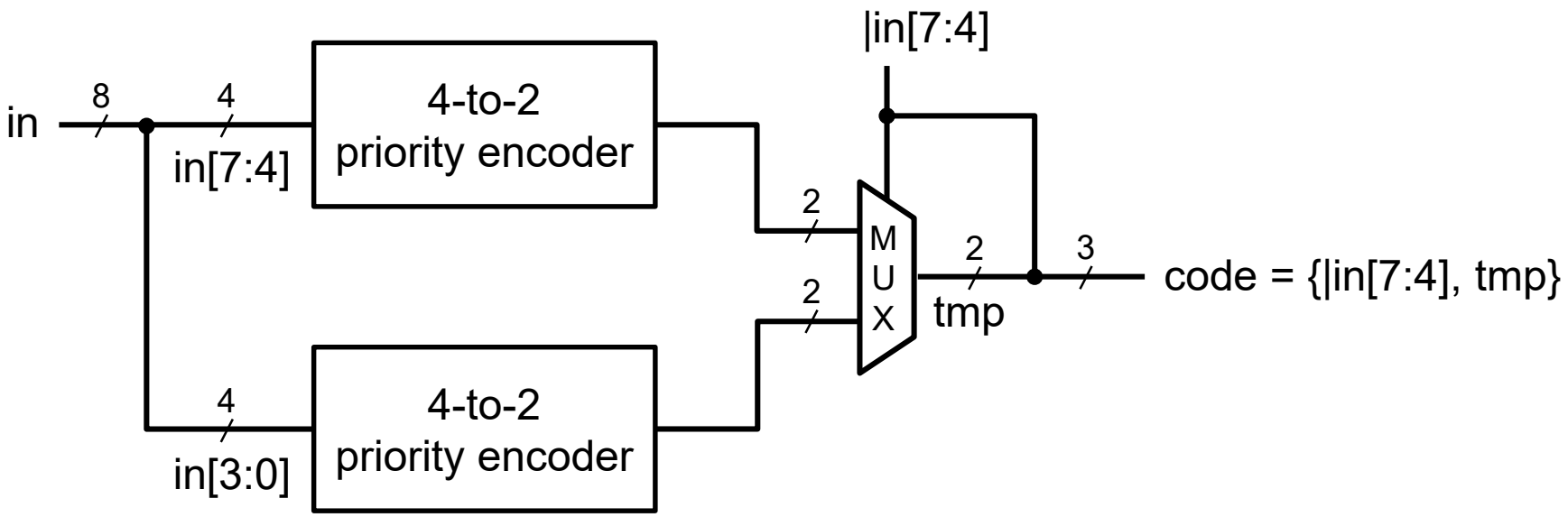
4-to-2 priority encoder
- 입력: in[7:4]
- 출력: code[2] = |in[7:4]

4-to-2 priority encoder
- 입력: in[3:0]
- 출력: code[2] = |in[3:0]

2-to-1 MUX

레이아웃 위한 설계

- 예: 4-to-2 priority encoder로 8-to-3 priority encoder



레이아웃 위한 설계

- 예: 4-to-2 priority encoder로 8-to-3 priority encoder

```
module pri_encoder83(
   input  [7:0] in,
   output [2:0] code
);

wire [1:0] code1, code2, tmp;

pri_encoder42 u1(.in(in[7:4]), .code(code1));
pri_encoder42 u2(.in(in[3:0]), .code(code2));

assign tmp = (|in[7:4]) ? code1 : code2;
assign code = {|in[7:4], tmp};

endmodule
```